School of Future Tech

Case Study Report

on

Stock Trading Simulation System

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1. Introduction to the Case Study

Stock trading is an important part of modern financial markets where investors buy and sell shares of companies to earn profits. Understanding the process of stock trading can be challenging for beginners because real-world trading involves financial risk and complex systems.

This case study focuses on designing and implementing a **Stock Trading Simulation System** that allows users to simulate stock trading activities without using real money. The system enables users to buy and sell stocks, maintain a portfolio, and track the overall value of their investments.

The case study demonstrates how programming concepts such as **classes, arrays, and functions in C++** can be used to build a simple trading system. It also highlights how simulation can help users understand the fundamentals of stock trading and portfolio management.

2. Problem Statement / Case Background (Abstract)

Background

Stock trading platforms allow investors to place buy and sell orders, track their investments, and calculate profits or losses based on market changes. However, beginners often lack practical experience with such systems.

A simulation system helps users understand the trading process without financial risk. It provides a controlled environment where users can practice buying and selling stocks and observe how their portfolio value changes.

Abstract

This case study presents the design and implementation of a **Stock Trading Simulation System using C++**. The system simulates basic stock trading operations such as viewing market prices, buying stocks, selling stocks, maintaining a portfolio, and calculating profit or loss.

The program uses a menu-driven interface that allows users to interact with the system through simple inputs. The system keeps track of the user's cash balance and the number of shares owned. The portfolio value is calculated by combining the current value of owned stocks with the remaining balance.

The implementation demonstrates how basic programming logic and object-oriented programming concepts can be used to simulate a simplified trading environment.

3. Case Study Design

The **Stock Trading Simulation System** is designed as a console-based application with the following components:

Market Data

A list of predefined stocks is stored in the system along with their prices.
Examples of stocks used in the simulation include:

- TCS
- INFY
- RELIANCE
- HDFC
- ADANI

Trading Operations

Users can perform trading actions such as:

- Viewing market prices
- Buying stocks
- Selling stocks

Portfolio Management

The system maintains a portfolio that records:

- The number of shares owned for each stock
- The user's remaining cash balance
- The total portfolio value

Profit and Loss Tracking

The system compares the current portfolio value with the initial investment amount to determine whether the user has made a profit or loss.

User Interaction

A menu-driven interface allows users to interact with the system by selecting options from the console.

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

Methods Used

Menu-driven program design
The system uses a menu-based interface where users select options to perform actions such as viewing market data or trading stocks.

Portfolio management logic
The program keeps track of stock quantities owned by the user and calculates the total value of holdings.

Profit and loss calculation
Profit or loss is calculated using the formula:

Current Portfolio Value = Cash Balance + Value of Owned Stocks

Profit/Loss = Current Portfolio Value − Initial Balance

Technology Stack Used for the Case

Programming Language

- C++

Libraries Used

- iostream (for input and output)
- string (for handling stock names)

Programming Concepts Applied

- Classes and Objects
- Arrays
- Functions
- Conditional statements
- Loops

Development Environment

- C++ Compiler (GCC / Dev C++ / Visual Studio Code)

Operating System

- Windows / Linux / macOS

5. Problem Statement / Case Study Implementation Details and Snapshots

The implementation of the system is divided into several components.

Stock Class

Stores information about stocks such as:

- Stock name
- Stock price

Portfolio Class

Manages the user's trading activities including:

- Cash balance
- Quantity of shares owned
- Buying stocks
- Selling stocks
- Portfolio display
- Profit and loss calculation

Main Program

The main program controls the interaction with the user through a menu-driven system.

The menu options include:

1. View Market Prices
2. Buy Stock
3. Sell Stock
4. View Portfolio
5. View Profit/Loss
6. Exit

Snapshots

Program Menu

```
===== Stock Trading Simulation =====
1. View Market
2. Buy Stock
3. Sell Stock
4. View Portfolio
5. View Profit/Loss
6. Exit
Enter choice:
```

Market Prices Display

```
--- Market Prices ---
1. TCS  Price: 3500
2. INFY  Price: 1500
3. RELIANCE  Price: 2700
4. HDFC  Price: 1600
5. ADANI  Price: 3000
```

Buy Stock

```
===== Stock Trading Simulation =====
1. View Market
2. Buy Stock
3. Sell Stock
4. View Portfolio
5. View Profit/Loss
6. Exit
Enter choice: 2
Enter stock number: 3
Enter quantity: 4
Stock purchased successfully.
```

Sell Stock

```
===== Stock Trading Simulation =====
1. View Market
2. Buy Stock
3. Sell Stock
4. View Portfolio
5. View Profit/Loss
6. Exit
Enter choice: 3
Enter stock number: 3
Enter quantity: 2
Stock sold successfully.
```

Portfolio

```
--- Portfolio ---
RELIANCE  Shares: 2  Value: 5400
Cash Balance: 94600
Total Portfolio Value: 100000
```

Profit/Loss

```
--- Profit / Loss ---
Initial Balance: 100000
Current Value: 100000
No Profit No Loss
```

Exit

```
===== Stock Trading Simulation =====
1. View Market
2. Buy Stock
3. Sell Stock
4. View Portfolio
5. View Profit/Loss
6. Exit
Enter choice: 6
Program ended.
```

6. Problem Statement / Case Study Results and Conclusion

Results

The system successfully simulates basic stock trading activities. Users can buy and sell stocks, track their portfolio value, and calculate profit or loss.

The simulation demonstrates how trading operations affect the user's portfolio balance and helps users understand how stock investments work.

Conclusion

This case study presents a **simple stock trading simulation system developed using C++**. The system allows users to practice trading in a simulated environment without financial risk.

By implementing features such as buying stocks, selling stocks, portfolio tracking, and profit/loss calculation, the project demonstrates how programming concepts can be applied to create a functional financial simulation system.

The project also provides a foundation for further enhancements such as real-time stock data integration, graphical interfaces, and advanced trading features.

7. References

C++ Programming Documentation for basic programming concepts and syntax.

Official documentation of standard C++ libraries such as **iostream** and **string**.

Online tutorials and learning resources related to **object-oriented programming in C++**.

Educational materials and articles related to **stock market trading and portfolio management**.